

THE EFFECT OF HIP JOINT ANGLE ON ISOMETRIC MIDTHIGH PULL KINETICS

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ABSTRACT

Dos'Santos, T, Thomas, C, Jones, PA, McMahon, JJ, and Comfort, P. The effect of hip joint angle on isometric midthigh pull kinetics. *J Strength Cond Res* 31(10): 2748–2757, 2017—The aim of this study were to compare isometric midthigh pull (IMTP) peak force (PF), time-specific force values (100-, 150-, and 200 ms), rate of force development (RFD) at predetermined time bands (0–100, 0–150, and 0–200 ms) and net forces between 2 commonly adopted hip joint angles (145° [hip₁₄₅] and 175° [hip₁₇₅]) with a 145° standardized knee angle. Twenty-eight collegiate athletes (age: 21.7 ± 1.5 years, height: 1.75 ± 0.08 m, mass: 81.5 ± 8.4 kg) performed 2 IMTP trials at each hip joint angle in a randomized counterbalanced order. A subgroup ($n = 10$) performed the IMTP testing 7 days later to establish the between-session reliability. Intra-class correlation coefficients (ICCs) and coefficient of variation (CV) demonstrated high within-session reliability and acceptable variability for all IMTP kinetics at each posture (ICC ≥ 0.86 , CV $\leq 13.7\%$), excluding hip₁₇₅ RFD 0–100 ms and net force at 100 ms which demonstrated greater variability (CV = 18.1–18.5%). High between-session reliability and acceptable variability were observed for all IMTP kinetics at each posture (ICC = 0.72–0.97, CV = 4.5–12.8%), excluding RFD 0–100 ms which demonstrated greater variability for both postures. Hip₁₄₅ produced significantly greater time-specific force values ($p \leq 0.025$, $g = 0.25$ – 0.28), RFD at predetermined time bands ($p \leq 0.001$, $g = 0.59$ – 0.78), and net forces ($p \leq 0.001$, $g = 0.57$ – 0.74) compared with hip₁₇₅. Trivial non-significant differences were demonstrated between postures for PF and force at 100 ms ($p > 0.05$, $g \leq 0.14$). Significantly greater body weights (weighing period force) were observed with hip₁₇₅ compared with hip₁₄₅ ($p < 0.001$, $g = 0.74$). Coaches should consider administering a hip₁₄₅ for IMTP

testing as greater IMTP kinetics and lower levels of pretension during the weighing period are achieved with this posture.

KEY WORDS rate of force development, time-specific force, peak force, net force, assessment

INTRODUCTION

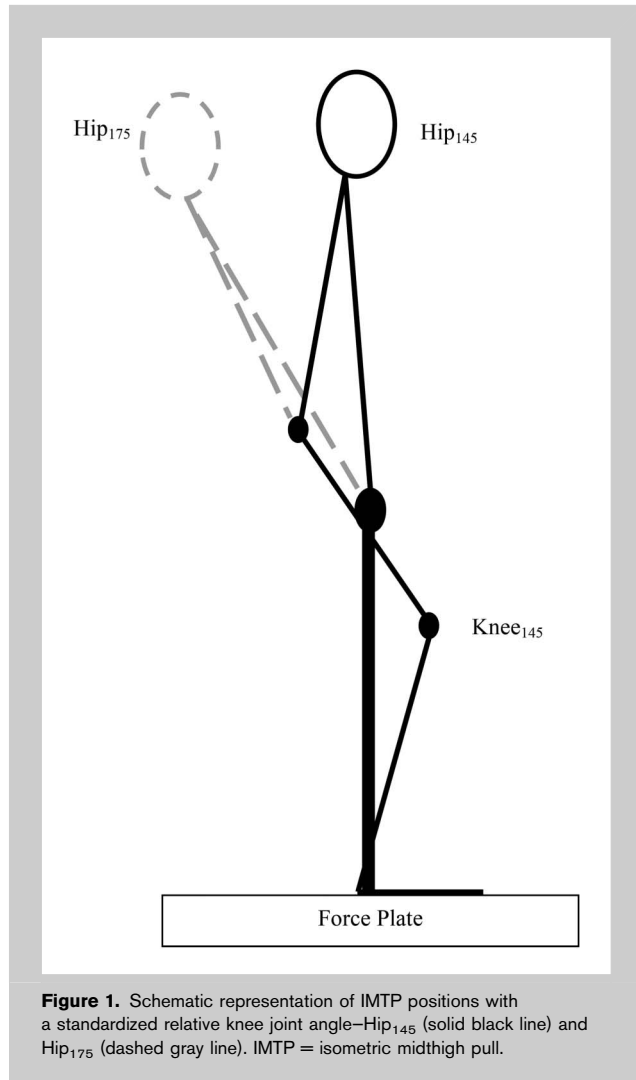
The neuromuscular qualities of the lower limb can be evaluated using force-time curves recorded from the isometric midthigh pull (IMTP). The IMTP is commonly used to assess the peak force (PF) production, but a further advantage is the ability to inspect an athlete's ability to produce force (4,12,15,22), rate of force development (RFD) (4,12,22), and impulse (8,43) at critical time intervals. The IMTP has been shown to demonstrate high within- and between-session reliability measures for PF (8,11,14,15,22), time-specific force values (4,12,15,22), RFD (12,22), and impulse (8,43) across different time intervals. Because of the high reliability and its simplicity to administer, the IMTP is commonly used to evaluate the force-time qualities to prescribe future training, such as inspection of the dynamic strength index when combined with PF during a jump (45), whereas it is also used to monitor adaptations to training (3). In addition, the IMTP can be used as a potentially safer surrogate to dynamic 1 repetition maximum (1RM) strength testing (8,11), with strong correlations observed between IMTP PF and 1RM back squat ($r \geq 0.96$) (33,35), snatch and clean and jerk ($r \geq 0.83$) (4), and deadlift ($r = 0.88$) (11). Moreover, inspecting neuromuscular preparedness (18) and assessment of bilateral (1) and unilateral force production asymmetries (13) are further purposes of IMTP testing.

The IMTP is modeled on the start of the second pull position of the clean whereby the largest forces, velocities, and power are generated (17,23). However, a contentious issue in IMTP testing is the selection of appropriate knee and hip joint angles. Currently, there is no agreed consensus on the appropriate knee and hip joint angles for IMTP testing. The IMTP was first introduced by Haff et al. (23), where knee and hip angles of $144 \pm 5^\circ$ and $145 \pm 3^\circ$ were reported, respectively. Since then, a diverse spectra of knee angles (120–145°) have been reported, including fixed specific

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angles of 130° (33,34) and 140° (9,36,38), whereas some researchers report a range of knee angles adopted by subjects, including 120–130° (10,46), 125 ± 5° (1), 125–135° (4), 127–145° (20,21,28), 137.6 ± 12.9° (22), 141 ± 10° (27), and 140–145° (40). Similarly, a wider range of hip joint angles have been reported within the literature ranging from flexed positions of 124 ± 11°, resulting in a clear forward lean of the trunk, upright positions of 140–145° (8,22,38,40), and more extended positions of 155–165° (39) and 170–175° (1,4,29,42,47). As such, these different knee and hip angles result in different body positions relative to the bar, which could impact the force production capabilities during IMTP testing (Figure 1) (5,6).

Notably, a large contingent of studies do not report their knee (12,13,30,41,43,45) or hip joint angles (12,13,20,21,28,30,33,34,36,37,44,45) for IMTP testing and simply describe the body positioning. For example, researchers have stated self-preferred position (12,13,43,45), bar position below crease of hip (36), bar position at height of knee

(30) (which is clearly not a midhigh pull, or the start of the second pull during a clean), upright trunk (10), near vertical trunk (40), shoulders placed over the bar (37), and flat trunk with shoulders in line with bar (46). Failure to provide hip and joint angles for IMTP testing makes determining and replicating IMTP protocols difficult. The inconsistencies in postures reported within the literature and failure to provide knee and joint angle data could result in discrepancies between studies; in particular, the reliability and range of correlations with dynamic performance. Of interest, some knee joint angles (127–145°) for IMTP testing were calculated during the 2-dimensional analysis of each subject's actual weightlifting performance (20,21), but hip joint angles were not provided. This method would require, firstly, athletes to be competent at the clean and, secondly, would require extensive periods to collect and analyze such data, and would therefore be too time consuming and impractical for testing large squads of athletes and cohorts of subjects.

Differences in joint angles can impact force production because of changes in the length tension relationship in skeletal muscle, whereas the force produced acts through different moment arms (48). Consequently, this can influence the contractile properties influencing force (5,6,32) and RFD (32) production during isometric testing. Marcora and Miller (32) reported differences in PF and maximum RFD during the isometric leg press between knee angles 120 and 90°, respectively. Similarly, Beckham et al. (5) compared isometric PF in key positions of the conventional deadlift (floor, knee, midhigh pull, and lockout) revealing that the midhigh pull position (knee angle 125°, hip angle 145°) generated the highest PF, significantly higher than any other position. However, the authors failed to provide the knee and hip joint angles of the lockout position, thus it is uncertain whether the hip and joint angles adopted for the lockout position were reflective of commonly adopted IMTP hip and knee joint angles reported within the literature.

There is a paucity of research comparing IMTP kinetics between different knee and hip joint angles. Comfort et al. (8) compared IMTP kinetics between commonly reported knee (120, 130, 140, and 150°) and hip angles (125 and 145°) and self-preferred posture reporting no meaningful or significant differences ($p > 0.05$, $d \leq 0.061$) in PF, maximal RFD, or impulse during 100-, 200-, and 300 ms across postures. In addition, high between-session reliability was observed for all kinetic variables irrespective of posture. The authors advocated the use of a self-preferred midhigh pull position for IMTP testing because of the high reliability and lack of differences with the other postures, with also the potential ability to speed up IMTP testing to a reduced learning effect. Contrary to the findings of Comfort et al. (8), Beckham et al. (6) has advocated a hip joint angle of approximately 145° for IMTP testing, reporting greater IMTP kinetics (PF and time-specific force values) compared with a flexed 125° hip joint angle (standardized 125° knee joint angle) with small-to-

large effect sizes in athletes with weightlifting experience and small-to-moderate effect sizes without weightlifting experience. As such, given the conflicting findings from these 2 studies, there is no consensus on the optimal joint angle for IMTP testing.

There is a requirement therefore for further investigations into the effects of different joint angles on IMTP kinetics, specifically comparing the commonly reported and adopted hip angles of 145° (8,22,23,38,40) and 175° (1,4,29,42,47) within the literature, which Comfort et al. (8) and Beckham et al. (6) did not investigate. These 2 hip angles result in different body positions relative to the bar, which could potentially effect force production during IMTP testing (Figure 1). Practitioners use the IMTP to assess the rapid force production properties of their athletes; thus, it is imperative that athletes adopt the most optimal and favorable position (joint angle) to rapidly produce force. Subsequently, the results from this study should provide greater insight into which positions are favorable and optimal for isometric rapid force production. Thus, the aim of this study were to compare IMTP PF, time-specific force values, RFD at predetermined time bands, and net forces between 2 different hip joint angles (145 and 175°) with a standardized knee angle of 145°. It was hypothesized that greater IMTP kinetics would be observed with a 145° hip joint angle compared with a 175° angle because of advantageous length tension relationships of the hip extensors.

METHODS

Experimental Approach to the Problem

A repeated measures, within-subjects design was used to evaluate the effects of hip joint angle (145 vs 175°) at a standardized knee joint angle (145°) on IMTP PF, time-specific force values, RFD at predetermined time bands, and net forces. A randomized and counterbalanced testing protocol was used to control for order effect whereby subjects performed 2 maximum effort IMTPs in each position while standing on a force plate sampling at 1,000 Hz. Isometric midhigh pull testing was performed on 2 separate testing sessions 7 days apart at the aforementioned postures to determine the within-session and between-session reliability of each measure and to determine the effect of hip joint angles on the dependent variables (PF, time-specific force values, RFD at predetermined time bands, and net forces).

Subjects

Twenty-eight (23 men and 5 women) collegiate athletes (age: 21.7 ± 1.5 years, height: 1.75 ± 0.08 m, mass: 81.5 ± 8.4 kg, relative one repetition maximum power clean: 1.06 ± 0.18 kg/BM) from rowing and soccer participated in this study. A subgroup ($n = 10$) returned on a second occasion 7 days later at the same time of the day to determine between-session reliability. Based on the work of Beckham et al. (5) for differences in isometric PF between postures, a minimum sample size of 15 was determined from an a priori power

analysis using G*Power (Version 3.1, University of Dusseldorf, Germany) (16) based on an effect size of 1.23, a power of 0.99, and type 1 error or alpha level of 0.05.

The investigation was approved by the University of Salford institutional ethics review board, and all subjects were informed of the benefits and risks of the investigation before signing an institutionally approved consent form to participate in the study. Subjects were familiar with the IMTP protocol and had ≥ 6 months resistance training experience of the power clean and its derivatives; all IMTP trials were assessed by certified strength and conditioning specialists. At the time of testing, subjects were at mid-season in the first week of a power mesocycle having performed a 4-week maximum strength mesocycle.

Procedures

All testing took place at the same time of day and a subgroup ($n = 10$) returned on a second occasion 7 days later at the same time of day to minimize the effect of circadian rhythm and to determine between-session reliability. Subjects were required to abstain from training for 48 hours before testing and asked to maintain a consistent fluid and dietary intake on each day of testing.

Preisometric Assessment Warm-up

All subjects performed a standardized warm-up comprising 10 body weight (BW) squats and lunges followed by 2 isometric efforts at a perceived intensity of 50, and 75% of maximum effort, interspersed with a 1-minute rest period (5,29).

Isometric Midhigh Pull Protocol

The IMTP testing was performed on a portable force plate sampling at 1,000 Hz (Model 9286AA, SN 1209740; Kistler, Winterthur, Switzerland) using a portable IMTP rack (Fitness Technology, Adelaide, Australia). Sampling at 1,000 Hz has been shown to produce high reliability for isometric force-time variables (12). A cold-rolled steel bar was positioned to correspond to the athlete's second pull power clean position where the bar height could be adjusted (3 cm increments) at various heights above the force plate to accommodate different sized athletes. Athletes were strapped to the bar in accordance with previous research (20) and positioned in 2 different postures; both postures required a standardized knee joint angle of 145°, however required different hip joint angles of 145° (Hip₁₄₅) and 175° (Hip₁₇₅), respectively. Subjects were placed in position; knee and hip relative angles (angle between 2 segments) were measured with goniometry to ensure that the position was accurately reproduced during each trial, with the bar resting midway up the thigh (approximately halfway between the iliac crest and the midpoint of the patella), just below the inguinal crease of the hip, to replicate the start position of the second pull phase of the clean. Hip₁₄₅ positioning resulted in a posture with shoulders directly above or slightly behind the bar; conversely, Hip₁₇₅ resulted

TABLE 1. Within-session reliability measures of IMTP kinetics across postures.*

Variable	ICC	95% CI		CV (%)	95% CI		SEM
		LB	UB		LB	UB	
Hip ₁₄₅	PF	0.99	0.97	1.00	2.8	1.8	68.1
	Force ₁₀₀	0.92	0.82	0.96	6.2	4.5	92.5
	RFD ₁₀₀	0.91	0.82	0.96	12.1	8.9	768.6
	Force ₁₅₀	0.97	0.94	0.99	4.6	3.4	72.0
	RFD ₁₅₀	0.97	0.93	0.98	7.5	4.9	461.5
	Force ₂₀₀	0.98	0.95	0.99	3.5	2.3	67.8
	RFD ₂₀₀	0.95	0.90	0.98	5.9	3.5	394.6
	Net PF	0.98	0.95	0.99	4.6	3.2	87.1
	Net Force ₁₀₀	0.91	0.81	0.96	11.6	8.6	80.4
	Net Force ₁₅₀	0.97	0.93	0.99	7.5	5.2	67.9
	Net Force ₂₀₀	0.96	0.91	0.98	5.9	3.8	74.8
	BW	0.95	0.90	0.98	3.5	2.1	35.4
	PF	0.99	0.98	1.00	2.8	2.1	66.2
	Force ₁₀₀	0.93	0.82	0.97	5.8	4.1	78.7
Hip ₁₇₅	RFD ₁₀₀	0.86	0.69	0.93	18.1	10.7	778.4
	Force ₁₅₀	0.95	0.88	0.98	5.3	3.9	83.5
	RFD ₁₅₀	0.90	0.78	0.95	13.3	8.4	2,004.3
	Force ₂₀₀	0.98	0.95	0.99	2.9	1.8	58.2
	RFD ₂₀₀	0.96	0.91	0.98	8.4	5.9	345.7
	Net PF	0.97	0.93	0.98	5.3	3.6	98.3
	Net Force ₁₀₀	0.83	0.64	0.92	18.5	11.7	86.0
	Net Force ₁₅₀	0.88	0.75	0.95	13.7	8.8	104.3
	Net Force ₂₀₀	0.95	0.89	0.98	7.5	4.8	75.1
	BW	0.93	0.85	0.97	5.0	3.1	53.2

*ICC = intraclass correlation coefficient; CI = confidence interval; CV = coefficient of variation; LB = lower bound; UB = upper bound; Hip₁₄₅ = hip joint angle 145°; PF = peak force; Force₁₀₀ = force at 100 ms; RFD = rate of force development; RFD₁₀₀ = RFD 0–100 ms; Force₁₅₀ = force at 150 ms; RFD₁₅₀ = RFD 0–150 ms; Force₂₀₀ = force at 200 ms; RFD₂₀₀ = RFD 0–200 ms; BW = bodyweight; Hip₁₇₅ = hip joint angle 175°.

in a posture with shoulders noticeably behind the bar as illustrated in Figure 1.

All subjects received standardized instructions to pull as fast and as hard as possible and push their feet directly into the force plate until being told to stop, as these instructions have been shown to produce optimal results (7). Once the body was stabilized (verified by watching the subject and force trace), the IMTP was initiated with the countdown “3, 2, 1 pull,” with subjects ensuring that maximal effort was applied for 5 seconds. Ground-reaction force data were collected for a duration of 8 seconds from the portable force platform that was interfaced with a laptop and recorded using Bioware software (Version 5.11; Kistler Instrument Corporation). Minimal pretension was allowed to ensure there was no slack in the body before initiation of pull, and subjects were instructed to be as still as possible during the weighing period, without initiating a pull on the bar, until given the instructions to “pull.” Trials without a stable baseline force trace (change in force >50 N) were rejected along with trials with a visible countermovements, subsequently another trial was performed (14,31). Subjects performed

a total of 2 maximal effort trials at each hip joint angle in a randomized and counterbalanced order, with each trial and interspersed with a 2-minute rest period. Strong verbal encouragement was given for all trials and subjects. In line with previous recommendations, if the difference between the 2 trials exceeded 250 N then a third trial was performed (4,29). The mean of 2 trials was used for statistical analyses.

Isometric Force-Time Curve Assessment

All force-time data recorded during the IMTP were inspected using a customized analysis Microsoft Excel spreadsheet (version 2016; Microsoft Corp., Redmond, WA, USA) to determine specific force-time characteristics. The maximum force generated during the 5 second maximum effort IMTP was reported as the absolute PF (22). In addition, time-specific force values at 100 ms (Force₁₀₀), 150 ms (Force₁₅₀), and 200 ms (Force₂₀₀) were calculated (12,22). Net PF and net time-specific force values were calculated by subtracting BW (calculated during 1 second weighing period) from the time-specific force value. Rate of force development at predetermined time bands 0–100, 0–150,

TABLE 2. Between-session reliability measures of IMTP kinetics across postures ($n = 10$).*

		Session 1		Session 2		ICC	95% CI			95% CI			<i>p</i>	<i>g</i>
		Mean	<i>SD</i>	Mean	<i>SD</i>		LB	UB	CV (%)	LB	UB	<i>SEM</i>		
Hip ₁₄₅	PF (N)	2,656.1	628.2	2,518.9	643.6	0.97	0.81	0.99	4.5	1.6	7.3	110.2	0.033	0.21
	Force ₁₀₀ (N)	1,370.3	280.0	1,381.3	267.2	0.84	0.33	0.96	8.0	3.6	12.5	109.5	0.872	−0.04
	RFD ₁₀₀ (N·s ^{−1})	5,548.4	2,552.3	5,729.5	1,802.2	0.85	0.36	0.96	15.2	9.4	21.1	867.0	0.740	−0.08
	Force ₁₅₀ (N)	1,731.0	381.0	1,690.8	394.3	0.90	0.61	0.98	7.6	3.5	11.7	121.4	0.608	0.10
	RFD ₁₅₀ (N·s ^{−1})	6,103.3	2,314.0	5,882.6	2,176.8	0.91	0.64	0.98	11.9	5.8	17.9	670.2	0.612	0.09
	Force ₂₀₀ (N)	1,960.3	425.0	1,883.7	437.3	0.94	0.77	0.99	6.6	4.0	9.1	106.5	0.264	0.17
	RFD ₂₀₀ (N·s ^{−1})	5,724.1	1,868.7	5,376.4	1,837.4	0.94	0.77	0.99	9.4	5.6	13.1	457.7	0.235	0.18
	Net PF (N)	1,883.5	577.5	1,746.9	585.1	0.96	0.79	0.99	6.6	2.2	11.1	114.8	0.050	0.23
	Net Force ₁₀₀ (N)	597.7	253.0	609.3	180.9	0.85	0.36	0.96	12.8	7.2	18.4	86.0	0.830	−0.05
	Net Force ₁₅₀ (N)	958.3	344.1	918.8	325.3	0.91	0.65	0.98	9.9	4.5	15.2	100.5	0.542	0.11
Hip ₁₇₅	Net Force ₂₀₀ (N)	1,187.7	371.0	1,111.7	366.1	0.94	0.76	0.99	9.7	6.3	13.1	91.8	0.191	0.20
	BW (N)	772.6	121.6	772.0	143.9	0.97	0.86	0.99	4.0	2.5	5.4	24.5	0.970	0.00
	PF (N)	2,556.4	611.2	2,531.8	574.6	0.97	0.89	0.99	5.3	3.0	7.6	101.0	0.716	0.04
	Force ₁₀₀ (N)	1,388.0	186.6	1,339.7	262.4	0.88	0.55	0.97	5.2	0.1	10.3	78.5	0.33	0.20
	RFD ₁₀₀ (N·s ^{−1})	4,484.1	1,665.4	4,247.9	1,327.8	0.68	−0.36	0.92	14.8	3.0	26.5	849.3	0.632	0.15
	Force ₁₅₀ (N)	1,663.4	278.3	1,593.2	339.0	0.93	0.73	0.98	5.1	0.7	9.5	82.1	0.175	0.22
	RFD ₁₅₀ (N·s ^{−1})	4,825.7	1,549.6	4,521.9	1,380.3	0.83	0.73	0.98	13.1	4.5	21.8	603.2	0.878	0.20
	Force ₂₀₀ (N)	1,913.4	375.7	1,839.3	390.2	0.95	0.80	0.99	5.3	1.7	9.0	86.5	0.181	0.19
	RFD ₂₀₀ (N·s ^{−1})	4,869.4	1,516.1	4,621.9	1,363.8	0.89	0.59	0.97	12.8	5.7	19.9	471.7	0.141	0.16
	Net PF (N)	1,660.46	551.72	1,655.81	476.37	0.95	0.80	0.99	9.5	4.3	14.7	112.9	0.951	0.01
	Net Force ₁₀₀ (N)	492.04	169.37	463.72	142.11	0.72	−0.15	0.93	11.6	0.9	22.4	82.4	0.561	0.17
	Net Force ₁₅₀ (N)	767.48	236.99	717.21	212.07	0.85	0.43	0.96	11.0	3.0	19.0	87.4	0.355	0.21
	Net Force ₂₀₀ (N)	1,017.52	307.14	963.32	275.49	0.89	0.608	0.974	11.9	5.7	18.0	93.6	0.364	0.18
	BW (N)	895.9	167.5	876.0	158.6	0.97	0.88	0.99	3.2	1.2	5.3	29.6	0.3	0.12

*ICC = intraclass correlation coefficient; CI = confidence interval; LB = lower bound; UB = upper bound; CV = coefficient of variation; Hip₁₄₅ = hip joint angle 145°; PF = peak force; Force₁₀₀ = force at 100 ms; RFD = rate of force development; RFD₁₀₀ = RFD 0–100 ms; Force₁₅₀ = force at 150 ms; RFD₁₅₀ = RFD 0–150 ms; Force₂₀₀ = force at 200 ms; RFD₂₀₀ = RFD 0–200 ms; BW = bodyweight; Hip₁₇₅ = hip joint angle 175°.

and 0–200 ms (RFD₁₀₀, RFD₁₅₀, and RFD₂₀₀) were also calculated using the equation: $RFD = \Delta \text{force} / \Delta \text{time interval}$ (4,12,22,29). The onset of the contraction was determined when vertical ground-reaction force deviated 5 SDs of BW (14). The combined residual force and BW were calculated as the average force over a 1 second stationary weighing period (in midthigh pull position posture) before the initiation of the IMTP (14).

Statistical Analyses

Statistical analyses were performed using SPSS software version 23 (SPSS, Chicago, IL, USA). Normality for all variables was confirmed using a Shapiro-Wilks test. Within-session reliability and between-session reliability were assessed through intraclass correlation coefficients (ICCs), 95% confidence intervals, coefficient of variation (CV) calculated as $SD / \text{mean} \times 100$ and SEM. Minimum acceptable reliability was determined with an ICC of >0.7 and a CV of $<15\%$ (2,22).

Paired sample *t*-tests and effect sizes were used to compare IMTP kinetics between sessions. Differences in IMTP kinetics between postures were assessed using paired sample *t*-tests, effect sizes, mean differences, and percentage differences. Effect sizes were calculated using the Hedges' *g* method (24) and interpreted using the Hopkins' scale (25). The criterion for significance was set at $p \leq 0.05$.

RESULTS

High within-session reliability was observed for hip₁₄₅ IMTP PF (ICC = 0.99, CV = 2.8%), time-specific force values (ICC = 0.92–0.98, CV = 3.5–6.2%), RFD at predetermined time bands (ICC = 0.91–0.97, CV = 5.9–12.1%), and net

forces (ICC = 0.91–0.98, CV = 4.6–11.6%)—all achieving minimum acceptable reliability criteria (Table 1). With the exception of hip₁₇₅ RFD₁₀₀ and net force₁₀₀, which failed to meet the minimum acceptable reliability criteria, high within-session reliability was observed for hip₁₇₅ IMTP PF (ICC = 0.99, CV = 2.8%), time-specific force values (ICC = 0.93–0.98, CV = 2.9–5.8%), RFD at predetermined time bands (ICC = 0.86–0.96, CV = 8.4–13.3%), and net forces (ICC = 0.83–0.97, CV = 5.3–13.7%) (Table 1). Body weight was highly reliable, irrespective of the posture (ICC = 0.93–0.95, CV = 3.5–5.0%) (Table 1).

Between-session testing demonstrated high reliability for all kinetics across both postures (ICC = 0.72–0.97, CV = 4.5–12.8%) with the exception of RFD₁₀₀ which failed to achieve the minimum acceptable reliability criteria for both postures (Table 2). Significant differences between sessions were observed only for hip₁₄₅ PF ($p = 0.033$) and net PF ($p = 0.05$) with effect sizes revealing a small difference ($g = 0.21$ – 0.23). No other significant differences ($p > 0.05$, $g \leq 0.22$) were observed between sessions for all IMTP kinetics across both postures (Table 2).

Isometric mid-thigh pull descriptive statistics between postures are presented in Table 3 along with *p* values, effect sizes, mean, and percentage differences. Trivial nonsignificant differences were demonstrated between postures for PF and force₁₀₀ ($p > 0.05$, $g \leq 0.14$). However, hip₁₄₅ produced significantly greater time-specific force values ($p \leq 0.025$, $g = 0.25$ – 0.28), RFD at predetermined time bands ($p \leq 0.001$, $g = 0.59$ – 0.78), and net forces ($p \leq 0.001$, $g = 0.57$ – 0.74) in comparison with hip₁₇₅, with effect sizes indicating small-to-moderate differences (Table 3). Conversely, significantly higher BW ($p < 0.001$) was observed with

TABLE 3. Comparisons of IMTP kinetics between postures.*

Variable	Hip ₁₄₅		Hip ₁₇₅		<i>p</i>	<i>g</i>	Descriptor	Mean		% Difference	SD
	Mean	SD	Mean	SD				difference	SD		
PF (N)	2,839.5	677.7	2,747.2	627.3	0.066	0.14	Trivial	92.3	255.4	2.6	8.9
Force ₁₀₀ (N)	1,483.6	306.5	1,488.2	284.9	0.893	−0.02	Trivial	−4.6	179.0	−1.3	12.9
RFD ₁₀₀ (N·s ^{−1})	6,095.4	2,516.5	4,766.4	1,908.0	0.001	0.59	Small	1,329.0	1,821.0	16.8	30.2
Force ₁₅₀ (N)	1,900.8	416.4	1,792.3	357.9	0.025	0.28	Small	108.5	241.4	4.5	13.1
RFD ₁₅₀ (N·s ^{−1})	6,844.5	2,426.4	5,204.7	1,639.9	<0.001	0.78	Moderate	1,639.9	1,710.1	21.1	24.5
Force ₂₀₀ (N)	2,125.5	442.2	2,017.5	398.0	0.022	0.25	Small	108.0	234.2	4.4	10.9
RFD ₂₀₀ (N·s ^{−1})	6,257.0	1,802.3	5,029.4	1,611.9	<0.001	0.71	Moderate	1,227.5	1,318.2	18.5	20.7
Net PF (N)	2,018.7	581.8	1,791.5	524.4	<0.001	0.40	Small	227.2	298.4	10.4	14.8
Net Force ₁₀₀ (N)	662.9	255.2	532.6	192.9	0.001	0.57	Small	130.3	189.3	14.4	30.6
Net Force ₁₅₀ (N)	1,080.0	367.5	836.6	286.8	<0.001	0.73	Moderate	243.4	263.5	19.5	24.0
Net Force ₂₀₀ (N)	1,304.7	367.3	1,061.8	320.8	<0.001	0.69	Moderate	242.9	265.7	17.3	19.4
BW (N)	820.8	157.7	955.7	201.3	<0.001	−0.74	Moderate	−134.9	71.5	−16.2	8.0

*Hip₁₄₅ = hip joint angle 145°; Hip₁₇₅ = hip joint angle 175°; PF = peak force; Force₁₀₀ = Force at 100 ms; RFD = rate of force development; RFD₁₀₀ = RFD 0–100 ms; Force₁₅₀ = force at 150 ms; RFD₁₅₀ = RFD 0–150 ms; Force₂₀₀ = Force at 200 ms; RFD₂₀₀ = RFD 0–200 ms; BW = bodyweight.

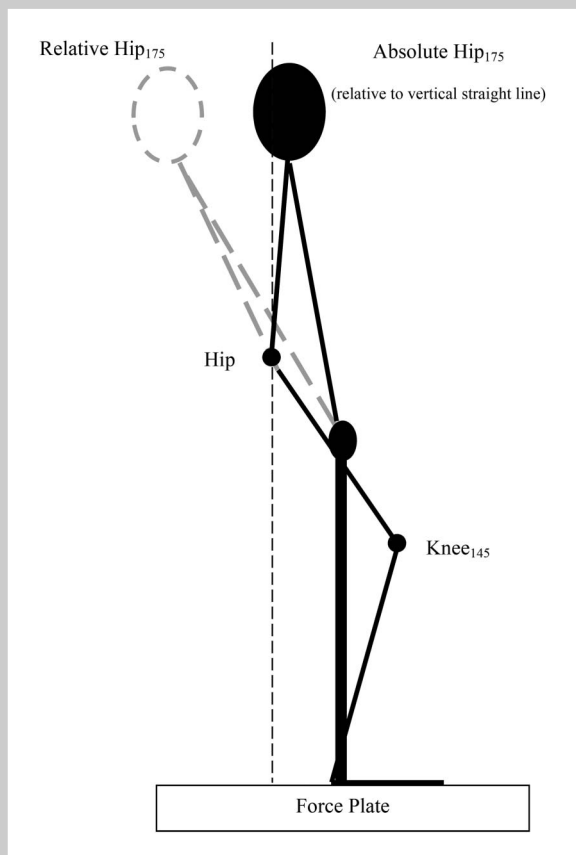


Figure 2. Schematic representation of relative and absolute 175° hip joint angles IMTP positions with a standardized relative knee joint angle—absolute (trunk) hip₁₇₅ (solid black line) and relative hip₁₇₅ (dashed gray line). IMTP = isometric midhigh pull.

a hip₁₇₅ angle compared with hip₁₄₅, with a moderate effect size ($g = 0.74$) (Table 3).

DISCUSSION

The aim of the present study was to compare IMTP kinetics between commonly reported hip joint angles 145° and 175° with a standardized knee joint angle. **This study is the first to compare a hip₁₇₅ joint angle to a hip₁₄₅ joint angle, finding significantly greater time-specific force values, RFD at predetermined time bands, and net forces with a hip₁₄₅ posture compared with a more extended hip₁₇₅ posture** (Table 3), in agreement with our hypotheses. Equally, both postures resulted in high within-session and between-session reliability measures for all IMTP kinetics, with the exception of RFD₁₀₀ that failed to meet minimum acceptable reliability for both postures between sessions (Tables 1 and 2). **Consequently, the results from this study suggest that a 145° hip angle is a more favorable position for rapid force production compared with a more extended hip position (175°) during IMTP testing.** Therefore, practitioners should consider administering IMTP testing with

an approximate 145° relative hip joint angle compared with a 175° hip joint angle, while also acknowledging that adopting different and inconsistent joint angles can significantly influence IMTP kinetics.

A diverse range of hip and knee joint angles have been reported within the literature for IMTP testing (1,4,22,23,38,39,42,47). To our knowledge, the studies by Comfort et al. (8) and Beckham et al. (6) are the only ones to examine the effect of joint angle on a range of IMTP kinetics reporting conflicting findings. Comfort et al. (8) demonstrated no significant or meaningful differences ($p > 0.05$, $d \leq 0.061$) for PF, maximum RFD, and impulse during 100-, 200-, and 300 ms between joint angles, and the authors advocated the use of a self-preferred midhigh pull position to minimize the learning effect. Conversely, Beckham et al. (6) found that greater PF and time-specific force values (small-to-large effect sizes) were achieved with a hip joint angle of 145° compared with a more flexed 125° angle. The present study compared a hip₁₄₅ joint angle to an extended hip₁₇₅ joint angle reporting no significant differences in PF between postures, but small-to-moderate significant differences in time-specific force values, RFD at predetermined time bands, and net forces were observed between postures (Table 3). Notably, greater mean and percentage differences were observed for net forces and RFD variables (Table 3) between postures indicating a greater influence on these kinetic variables. As such, the results from the present study are in agreement with Beckham et al. (6), highlighting that hip joint angle and subsequent body position influence isometric rapid force production. Supporting the recommendations of Beckham et al. (6), we recommend coaches and researchers should consider administering an approximate 145° hip joint angle for IMTP testing.

The data from the present study show that a hip₁₄₅ position appears to be a favorable position to assess the rapid force production capabilities of athletes while also demonstrating that hip joint angle directly influences time-specific force, net force, and RFD characteristics (Table 3). This is supported by previous studies that have shown differences in maximum RFD and PF between 90° and 120° knee flexion during the isometric leg press (32). Beckham et al. (5) also observed significant differences in PF between various positions in the deadlift and the midhigh pull positions. Of interest, the authors compared a midhigh pull position to a deadlift lockout position demonstrating large differences in PF ($d = 1.23$); however, PF was the only kinetic variable examined and the specific joint angles of the lockout position were not provided. Nonetheless, based on the results of this study and corroborative research, lower limb joint angle influences force production during isometric testing (5,6,32). As such, coaches and researchers should ensure joint angles are standardized and consistent between testing occasions to allow valid comparisons of performance variables when longitudinally monitoring neuromuscular performance, so such changes in IMTP kinetics can be

attributed to training or fatigue, and not to differences in joint angles.

Coaches use the IMTP to assess the rapid force production properties of their athletes to monitor and inform future training; thus, it is imperative that athletes adopt the most optimal and favorable position (joint angle) to rapidly produce force. The results of the present study demonstrate that an extended hip joint angle of 175° was a suboptimal position for force production compared with hip₁₄₅ joint angle (Table 3), whereas Beckham et al. (6) observed that a flexed 125° hip joint angle was also suboptimal in force production compared with hip₁₄₅. Collectively, the results of these studies suggest that body position relative to the bar does matter for IMTP force production. Failure to place athletes in the optimal joint angles (body position) of hip₁₄₅ could limit rapid force production, potentially leading to misinterpretations of their force production capabilities.

A stable baseline force during the weighing period with minimal pretension before the onset of a rapid contraction is recommended when conducting isometric testing (31). Of interest, considerably greater BW (weighing period forces) was observed for the hip₁₇₅ posture compared with the hip₁₄₅ (Table 3). This indicates that higher levels of pretension were achieved with the hip₁₇₅ posture which is suboptimal for evaluating RFD during isometric testing (31), and should therefore be avoided for IMTP testing.

Notably, significantly greater RFD at predetermined time bands were demonstrated with a hip₁₄₅ posture compared with hip₁₇₅, with mean percentage differences ranging from 16.8 to 21.1% (Table 3). As RFD was calculated as $\Delta\text{force} / \Delta\text{time interval}$, the consistently greater RFD may be explained by several factors, including the significantly greater net forces and lower BW (weighing period forces), which have a direct effect on the change in force component of the RFD equation. In addition, significantly greater force₁₅₀ and force₂₀₀ values were observed with a hip₁₄₅ joint angle directly influencing RFD. Collectively, the abovementioned factors, such as lower BW (weighing period force to determine the onset of contraction—due to lower pretension), greater net forces, and time-specific forces, result in a greater change in force, thus greater RFD with a hip₁₄₅ joint angle. Thus, practitioners are recommended to administer IMTP testing with a hip₁₄₅ joint angle for a more favorable position to attain RFD, time-specific force values, and net force data.

Numerous investigations have adopted hip joint angles of approximately 170–175° during IMTP testing (1,4,29,42,47); however, interpretation of these aforementioned studies may be limited; because the results of the present study indicate higher levels of pretension, lower RFD, lower time-specific forces, lower net forces, and lower reliability measures are achieved with the hip₁₇₅ posture (Tables 1–3). This posture appears to be a suboptimal position for producing force and RFD compared with hip₁₄₅, potentially because of differences in length-tension relationship of the hip extensors and differences in moment arms (48). We suggest that the hip

joint angles reported in the aforementioned studies (1,4,29,42,47) may be misrepresented and we question whether the authors are potentially referring absolute hip or trunk angle relative to a vertical straight line in comparison with measuring relative joint angle (angle between 2 segments meeting at a point) as performed in the present study (19) (Figure 2). Figure 2 illustrates the notable differences in trunk position relative to the bar between absolute and relative hip₁₇₅ joint angles. Therefore, coaches and researchers are advised to specify and standardize their knee and hip joint angles adopted for IMTP testing and state whether absolute or relative joint angles were measured to avoid confusion and allow the replication of IMTP testing methodologies.

The present study found PF to demonstrate the highest between-session reliability measures for both postures (Table 2) similar to the observations of previous between-session (ICC ≥ 0.89 , CV $\leq 4.6\%$) (8,11,15,45) and within-session research (ICC ≥ 0.97 , CV $\leq 3.2\%$) (12,22). Equally, both postures demonstrated high levels of within-session reliability for time-specific force values (Table 1) comparable with the reliability measures reported in previous research (4,12,22,29). Limited studies have inspected the between-session reliability of time-specific force values (15,26). High and acceptable between-session reliability measures were demonstrated for all time-specific force values (Table 2) in accordance with the reliability measures reported in youth male soccer players (15) and higher than the measures reported by James et al. (26). The results from this study confirm that both postures produce equally high within-session and between-session reliability measures for PF and time-specific force values.

Haff et al. (22) has shown that the method to quantify RFD can influence the resultant value and reliability of such measures, and as such, using predetermined time bands to calculate RFD has been recommended. To our knowledge, only 1 other study (26) has assessed the between-session reliability of RFD at predetermined time bands. RFD₁₀₀ in the present exceeded the minimum acceptable reliability criteria at both postures (between sessions) similar to the results of previous research (26). Conversely, lower and acceptable levels of variance were demonstrated for RFD₁₅₀ and RFD₂₀₀, consistent with the results of James et al. (26) who also showed improved RFD reliability measures over longer time intervals. High and acceptable within-session reliability measures were observed for all RFD variables during hip₁₄₅ testing; however, RFD₁₀₀ and net force₁₀₀ exceeded minimum acceptable reliability at hip₁₇₅ posture (Table 1). Therefore, the results from this study confirm that a hip₁₄₅ posture produces high within-session reliability for all RFD variables; however, both postures result in unacceptable reliability for RFD₁₀₀ (Tables 1 and 2).

It should be acknowledged that the present study only examined the effect of 2 different hip joint angles (145 and 175°) on IMTP kinetics, whereas Beckham et al. (6) only

compared 2 hip joint angles (145 and 125°) as well. Comfort et al. (8) recommends the use of a self-preferred selection of knee and hip joint angles as no significant differences between self-preferred position and a range of knee (120, 130, 140, and 150°) and hip joint angles were observed. The present study and Beckham et al. (6) both demonstrated greater force production with a hip₁₄₅ but did not compare this to a self-preferred position. Therefore, further research is required comparing hip₁₄₅ joint angle to self-preferred position to determine which body position results in optimal force production.

PRACTICAL APPLICATIONS

Coaches and researchers should conduct IMTP testing with a 145° hip joint angle because of the greater IMTP kinetics and lower levels of pretension observed in this position compared with a 175° hip joint angle. As such, coaches and researchers should ensure that joint angles are standardized and kept consistent between testing occasions to allow valid comparisons of performance variables when longitudinally monitoring neuromuscular performance, so such changes in IMTP kinetics can be attributed to training or fatigue, and not to differences in joint angles. Furthermore, researchers are recommended when publishing research to report the knee and hip joint angles adopted for IMTP testing because of the effect on IMTP kinetics and reliability, while specifying if relative or absolute joint angles were measured.

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REFERENCES

1. Bailey, C, Sato, K, Alexander, R, Chiang, CY, and Stone, MH. Isometric force production symmetry and jumping performance in collegiate athletes. *J Trainol* 2: 1–5, 2013.
2. Baumgartner, TA and Chung, H. Confidence limits for intraclass reliability coefficients. *Meas Phys Educ Exerc Sci* 5: 179–188, 2001.
3. Beattie, K, Carson, BP, Lyons, M, and Kenny, IC. The effect of maximal and explosive strength training on performance indicators in cyclists. *Int J Sports Physiol Perform* 12: 470–480, 2017.
4. Beckham, G, Mizuguchi, S, Carter, C, Sato, K, Ramsey, M, Lamont, H, Hornsby, G, Haff, G, and Stone, M. Relationships of isometric mid-thigh pull variables to weightlifting performance. *J Sports Med Phys Fitness* 53: 573–581, 2013.
5. Beckham, G, Lamont, HS, Sato, K, Ramsey, MW, and Stone, MH. Isometric strength of powerlifters in key positions of the conventional deadlift. *J Trainol* 1: 32–35, 2012.
6. Beckham, GK, Sato, K, Mizuguchi, S, Haff, GG, and Stone, MH. Effect of body position on force production during the isometric mid-thigh pull. *J Strength Cond Res* 2017. Epub ahead of print.
7. Bembien, MG, Clasey, JL, and Massey, BH. The effect of the rate of muscle contraction on the force-time curve parameters of male and female subjects. *Res Q Exerc Sport* 61: 96–99, 1990.
8. Comfort, P, Jones, PA, McMahon, JJ, and Newton, R. Effect of knee and trunk angle on kinetic variables during the isometric mid-thigh pull: Test-retest reliability. *Int J Sports Physiol Perform* 10: 58–63, 2015.
9. Cormie, P, McCauley, GO, Triplett, NT, and McBride, JM. Optimal loading for maximal power output during lower-body resistance exercises. *Med Sci Sports Exerc* 39: 340, 2007.
10. Darrall-Jones, JD, Jones, B, and Till, K. Anthropometric and physical profiles of English academy rugby union players. *J Strength Cond Res* 29: 2086–2096, 2015.
11. De Witt, JK, English, KL, Crowell, JB, Kalogera, KL, Williams, ME, Nieschwitz, BE, Hanson, AM, and Ploutz-Snyder, LL. Isometric mid-thigh pull reliability and relationship to deadlift 1RM. *J Strength Cond Res* 2016. Epub ahead of print.
12. Dos Santos, T, Jones, PA, Kelly, J, McMahon, JJ, Comfort, P, and Thomas, C. Effect of sampling frequency on isometric mid-thigh pull kinetics. *Int J Sports Physiol Perform* 11: 255–260, 2016.
13. Dos Santos, T, Thomas, C, Jones, PA, and Comfort, P. Assessing muscle strength asymmetry via a unilateral stance isometric mid-thigh pull. *Int J Sports Physiol Perform* 12: 505–511, 2017.
14. Dos Santos, T, Jones, PA, Comfort, P, and Thomas, C. Effect of different onset thresholds on isometric mid-thigh pull force-time variables. *J Strength Cond Res* 2016. Epub ahead of print.
15. Dos Santos, T, Thomas, C, Comfort, P, McMahon, JJ, Jones, PA, Oakley, NP, and Young, AL. Between-session reliability of isometric mid-thigh pull kinetics and maximal power clean performance in male youth soccer players. *J Strength Cond Res*, 2017.
16. Faul, F, Erdfelder, E, Buchner, A, and Lang, AG. Statistical power analyses using G* Power 3.1: Tests for correlation and regression analyses. *Behav Res Methods* 41: 1149–1160, 2009.
17. Garhammer, J. A review of power output studies of olympic and Powerlifting: Methodology, performance prediction, and evaluation tests. *J Strength Cond Res* 7: 76–89, 1993.
18. Gescheit, DT, Cormack, SJ, Reid, M, and Duffield, R. Consecutive days of prolonged tennis match play: Performance, physical, and perceptual responses in trained players. *Int J Sports Physiol Perform* 10, 2015.
19. Grimshaw, P, Fowler, N, Lees, A, and Burden, A. *Sport and Exercise Biomechanics (BIOS Instant Notes)*. Abingdon, United Kingdom: Taylor and Francis; 2004.
20. Haff, GG, Carlock, JM, Hartman, MJ, Kilgore, JL, Kawamori, N, Jackson, JR, Morris, RT, Sands, WA, and Stone, MH. Force-time curve characteristics of dynamic and isometric muscle actions of elite women olympic weightlifters. *J Strength Cond Res* 19: 741–748, 2005.
21. Haff, GG, Jackson, JR, Kawamori, N, Carlock, JM, Hartman, MJ, Kilgore, JL, Morris, RT, Ramsey, MW, Sands, WA, and Stone, MH. Force-time curve characteristics and hormonal alterations during an eleven-week training period in elite women weightlifters. *J Strength Cond Res* 22: 433–446, 2008.
22. Haff, GG, Ruben, RP, Lider, J, Twine, C, and Cormie, P. A comparison of methods for determining the rate of force development during isometric mid-thigh clean pulls. *J Strength Cond Res* 29: 386–395, 2015.
23. Haff, GG, Stone, M, O'Bryant, HS, Harman, E, Dinan, C, Johnson, R, and Han, KH. Force-time dependent characteristics of dynamic and isometric muscle actions. *J Strength Cond Res* 11: 269–272, 1997.
24. Hedges, L and Olkin, I. *Statistical Methods for Meta-analysis*. New York, NY: Academic Press, 1985.
25. Hopkins, WG. A scale of magnitudes for effect statistics. A new view of statistics, 2002. Available at: <http://www.sportsci.org/resource/stats/index.html>.
26. James, LP, Roberts, LA, Haff, GG, Kelly, VG, and Beckman, EM. The validity and reliability of a portable isometric mid-thigh clean pull. *J Strength Cond Res* 31: 1378–1386, 2017.
27. Kawamori, N, Rossi, SJ, Justice, BD, Haff, EE, Pistilli, EE, O'Bryant, HS, Stone, MH, and Haff, GG. Peak force and rate of force development during isometric and dynamic mid-thigh clean pulls performed at various intensities. *J Strength Cond Res* 20: 483–491, 2006.

28. Khamoui, AV, Brown, LE, Nguyen, D, Uribe, BP, Coburn, JW, Noffal, GJ, and Tran, T. Relationship between force-time and velocity-time characteristics of dynamic and isometric muscle actions. *J Strength Cond Res* 25: 198–204, 2011.
29. Kraska, JM, Ramsey, MW, Haff, GG, Fethke, N, Sands, WA, Stone, ME, and Stone, MH. Relationship between strength characteristics and unweighted and weighted vertical jump height. *Int J Sports Physiol Perform* 4: 461–473, 2009.
30. Lawton, TW, Cronin, JB, and McGuigan, MR. Anthropometry, strength and benchmarks for development: A basis for junior rowers' selection? *J Sports Sci* 30: 995–1001, 2012.
31. Maffiuletti, NA, Aagaard, P, Blazevich, AJ, Folland, J, Tillin, N, and Duchateau, J. Rate of force development: Physiological and methodological considerations. *Eur J Appl Physiol* 116: 1–26, 2016.
32. Marcora, S and Miller, MK. The effect of knee angle on the external validity of isometric measures of lower body neuromuscular function. *J Sports Sci* 18: 313–319, 2000.
33. McGuigan, MR, Newton, MJ, Winchester, JB, and Nelson, AG. Relationship between isometric and dynamic strength in recreationally trained men. *J Strength Cond Res* 24: 2570–2573, 2010.
34. McGuigan, MR and Winchester, JB. The relationship between isometric and dynamic strength in college football players. *J Sports Sci Med* 7: 101, 2008.
35. McGuigan, MR, Winchester, JB, and Erickson, T. The importance of isometric maximum strength in college wrestlers. *J Sports Sci Med* 5: 108–113, 2006.
36. Nuzzo, JL, McBride, JM, Cormie, P, and McCaulley, GO. Relationship between countermovement jump performance and multijoint isometric and dynamic tests of strength. *J Strength Cond Res* 22: 699–707, 2008.
37. Secomb, JL, Lundgren, LE, Farley, OR, Tran, TT, Nimphius, S, and Sheppard, JM. Relationships between lower-body muscle structure and lower-body strength, power, and muscle-tendon complex stiffness. *J Strength Cond Res* 29: 2221–2228, 2015.
38. Spiteri, T, Nimphius, S, Hart, NH, Specos, C, Sheppard, JM, and Newton, RU. The contribution of strength characteristics to change of direction and agility performance in female basketball athletes. *J Strength Cond Res* 28: 2415–2423, 2014.
39. Stone, MH, Sanborn, K, O'Bryant, HS, Hartman, M, Stone, ME, Proulx, C, Ward, B, and Hruby, J. Maximum strength-power-performance relationships in collegiate throwers. *J Strength Cond Res* 17: 739–745, 2003.
40. Stone, MH, Sands, WA, Carlock, JON, Callan, SAM, Dickie, DES, Daigle, K, Cotton, J, Smith, SL, and Hartman, M. The importance of isometric maximum strength and peak rate-of-force development in sprint cycling. *J Strength Cond Res* 18: 878–884, 2004.
41. Stone, MH, Sands, WA, Pierce, KC, Carlock, JON, Cardinale, M, and Newton, RU. Relationship of maximum strength to weightlifting performance. *Med Sci Sports Exerc* 37: 1037–1043, 2005.
42. Stone, MH, Sands, WA, Pierce, KC, Ramsey, MW, and Haff, GG. Power and power potentiation among strength-power athletes: Preliminary study. *Int J Sports Physiol Perform* 3: 55–67, 2008.
43. Thomas, C, Comfort, P, Chiang, CY, and Jones, P. Relationship between isometric mid-thigh pull variables and sprint and change of direction performance in collegiate athletes. *J Trainol* 4: 6–10, 2015.
44. Thomas, C, Dos'Santos, T, Comfort, P, and Jones, PA. Relationship between isometric strength, sprint, and change of direction speed in male academy cricketers. *J Trainol* 5: 18–23, 2016.
45. Thomas, C, Jones, PA, and Comfort, P. Reliability of the dynamic strength index in collegiate athletes. *Int J Sports Physiol Perform* 10: 542–545, 2015.
46. West, DJ, Owen, NJ, Jones, MR, Bracken, RM, Cook, CJ, Cunningham, DJ, Shearer, DA, Finn, CV, Newton, RU, and Crewther, BT. Relationships between force time characteristics of the isometric midthigh pull and dynamic performance in professional rugby league players. *J Strength Cond Res* 25: 3070–3075, 2011.
47. Whittington, J, Schoen, E, Labounty, L, Hamdy, R, Ramsey, M, Stone, M, Sands, W, Haff, G, and Stone, M. Bone mineral density and content of collegiate throwers: Influence of maximum strength. *J Sports Med Phys Fitness* 49: 464, 2009.
48. Zatsiorsky, VM and Kraemer, WJ. *Science and Practice of Strength Training*. Champaign, IL: Human Kinetics, 2006.